Section 1

This section of the exam contains 11 single answer multiple choice questions. You should allow 30 – 35 minutes for this part. Attempt all questions.

Question 1 **Points: 1**

Consider the following system of equations over $\mathbb{Z}_7$,

$$\begin{array}{rcrcrcl} x & + & y & + & z & = & 5 \\ & - & y & + & 4z & = & 3 \\ x & + & 2y & + & 4z & = & 2 \end{array}$$

How many solutions does this system have?

(A) 7

(B) 1

(C) 0

(D) 7^2

(E) infinitely many

Question 2 **Points: 1**

Given are the permutations

$$\alpha = (1\ 4)\ (3\ 2\ 5) \qquad \text{and} \qquad \gamma = (1\ 4\ 3)\ (2\ 5)$$

of $\{1, 2, 3, 4, 5\}$ expressed in cyclic notation. Consider the equation

$$\gamma = \beta^{-1}\alpha\beta.$$

Which **one** of the following expressions for β solves this equation?

(A) $\beta = (1\ 2\ 5\ 3\ 4)$

(B) $\beta = (2\ 3\ 4)$

(C) $\beta = (1\ 2\ 3)\ (4\ 5)$

(D) $\beta = (1\ 2)\ (4\ 5)$

(E) $\beta = (1\ 2)$

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Question 3 **Points: 1**

Consider the matrix

$$M = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

with entries in $\mathbb{Z}_2$. Which **one** of the following is row equivalent to M and in reduced row echelon form?

(A) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

(E) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

Question 4 **Points: 1**

Given is the matrix $M = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}$ over the finite field $\mathbb{Z}_{11}$. Compute the characteristic polynomial of M.

(A) $\chi(\lambda) = \lambda^3 + 8\lambda^2 + 7\lambda + 4$

(B) $\chi(\lambda) = \lambda^3 + 2\lambda^2 + 3\lambda + 4$

(C) $\chi(\lambda) = (1 - \lambda)^3$

(D) $\chi(\lambda) = 10\lambda^3 + 2\lambda^2 + 3\lambda + 5$

(E) $\chi(\lambda) = \lambda^3 + 5\lambda^2 + 7\lambda + 9$

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Question 5 **Points: 1**

Find the steady state probability vector of the 3×3 regular stochastic matrix

$$M = \begin{bmatrix} 0 & \frac{2}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & 0 \end{bmatrix}$$

(A) $\frac{1}{3} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(B) $\frac{1}{10} \cdot \begin{bmatrix} 4 \\ 3 \\ 3 \end{bmatrix}$

(C) $\frac{1}{6} \cdot \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

(D) $\begin{bmatrix} 14 \\ 12 \\ 15 \end{bmatrix}$

(E) $\frac{1}{6} \cdot \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}$

Question 6 **Points: 1**

Given is the linear transformation $L : \mathbb{Z}_5^2 \to \mathbb{Z}_5^2$ defined by $L(x, y) = (2x - 2y, x - 2y)$. Which **one** of the following is the inverse of L?

(A) $L^{-1}(x, y) = (x - y, 3x - y)$

(B) $L^{-1}(x, y) = (x + y, 3x - y)$

(C) $L^{-1}(x, y) = (x - y, x + 3y)$

(D) $L^{-1}(x, y) = (x + y, x + 3y)$

(E) The function is not invertible

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Question 7 **Points: 1**

Consider the basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ of $\mathbb{Z}_2^3$, and let $v = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Give the correct expression for $[v]_{\mathcal{B}}$.

(A) $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

(B) $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

(C) $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

(E) $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

Question 8 **Points: 1**

Let $B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $D = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$, both of which are ordered bases for $\mathbb{R}^2$. Let $M = [\mathrm{id}]_D^B$ be the change of basis matrix from the basis B to the basis D. Which **one** of the following is true?

(A) $M = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$

(B) $M = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

(C) $M = \begin{bmatrix} \frac{1}{2} & 1 \\ 1 & \frac{1}{3} \end{bmatrix}$

(D) $M = -\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

(E) $M = \frac{1}{6} \begin{bmatrix} -3 & 1 \\ 1 & -2 \end{bmatrix}$

Question 9 **Points: 1**

Let M be the following matrix with entries in $\mathbb{Q}$:

$$\begin{bmatrix} 2 & 3 & 0 & 1 & 4 \\ 2 & -1 & 2 & 3 & 2 \\ 4 & -6 & 6 & 8 & 2 \end{bmatrix}$$

Which **one** of the following is correct:

(A) $\operatorname{rank}(M) = 2$ and $\operatorname{nullity}(M) = 3$.

(B) $\operatorname{rank}(M) = 3$ and $\operatorname{nullity}(M) = 2$.

(C) $\operatorname{rank}(M) = 1$ and $\operatorname{nullity}(M) = 2$.

(D) $\operatorname{rank}(M) = 2$ and $\operatorname{nullity}(M) = 1$.

(E) The rank and nullity of M are not defined.

Section 2

This section of the exam contains 21 multiple answer questions. Partial credit can be earned on these questions. You should allow 90 minutes for this part. Attempt all questions.

Question 10 **Points: 1**

Given is the matrix $M = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ 0 & 4 & 3 \end{bmatrix}$ over $\mathbb{R}$. Which of the following are eigenvectors of M? One or multiple options can be correct.

(A) $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

(C) $\begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$

(D) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(E) $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$

Question 11 **Points: 1**

Which of the following are groups with respect to the specified operation. One or multiple options can be correct.

(A) The set of odd permutations of $\mathrm{Sym}(n)$, under composition

(B) The symmetries of a regular hexagon, under composition

(C) The set of invertible matrices with entries in $\mathbb{Z}_3$, under matrix multiplication

(D) The set of non-zero integers, under multiplication

(E) The set of complex numbers, under multiplication

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Question 12 **Points: 1**

Consider the equation $6x + 1 = 2$. In which of the following is there a solution to this equation? One or multiple options can be correct.

(A) $\mathbb{Q}$

(B) $\mathbb{Z}_{12}$

(C) $\mathbb{C}$

(D) $\mathbb{Z}_{10}$

(E) $\mathbb{Z}_5$

Question 13 **Points: 1**

Which of the following elements do not have a multiplicative inverse in $\mathbb{Z}_{12}$? One or multiple options can be correct.

(A) 2

(B) 3

(C) 4

(D) 5

(E) 6

Question 14 **Points: 1**

Given is the matrix

$$M = \begin{pmatrix} 1 & 1 & -\lambda \\ \lambda & 0 & 2 \\ 1 & \lambda + 1 & 1 \end{pmatrix}$$

with entries in some finite field $\mathbb{Z}_n$, $n > 2$.
For which values of λ is M invertible over $\mathbb{Z}_5$, but not invertible over $\mathbb{Z}_3$? One or multiple options can be correct.

(A) $\lambda = 0$

(B) $\lambda = 1$

(C) $\lambda = 2$

(D) $\lambda = 3$

(E) $\lambda = 4$

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Question 15 **Points: 1**

Which of the following groups are cyclic? One or multiple options can be correct.

(A) $\mathrm{Alt}(3)$

(B) $\mathrm{Sym}(4)$

(C) $\langle(1\ 2\ 3\ 5\ 4)\rangle$

(D) $\langle(1\ 2\ 3\ 4\ 5\ 6), (1\ 3\ 5)\ (2\ 4\ 6)\rangle$

(E) $\mathbb{Z}_9 \times \mathbb{Z}_3$

Question 16 **Points: 1**

For which of the following groups under addition with given element g, is g a generator? One or multiple options can be correct.

(A) $\mathbb{Z}_5 \times \mathbb{Z}_4$, $g = (1, 1)$

(B) $\mathbb{Z}_{20} \times \mathbb{Z}_{10}$, $g = (4, 2)$

(C) $\mathbb{C}$, $g = 1$

(D) $\mathbb{Z}_{221}$, $g = 8$

(E) $\mathbb{Z}_3 \times \mathbb{Z}_3$, $g = (1, 1)$

Question 17 **Points: 1**

Which of the following statements about determinants of matrices over $\mathbb{R}$ are correct? One or multiple options can be correct.

(A) $\det\left(\begin{bmatrix} 3 & 5 \\ 2 & 3 \end{bmatrix}\right) = -1$

(B) $\det\left(4 \cdot \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\right) = 64$

(C) $\det\left(\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}\right) = 30$

(D) $\det\left(\begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}\right) = 16$

(E) $\det\left(\begin{bmatrix} 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \end{bmatrix}\right) = 16$

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Question 18 **Points: 1**

Which of the following subsets of $\mathbb{R}^3$ are subspaces? One or multiple options can be correct.

(A) $\{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$

(B) $\{(x, y, z) \in \mathbb{R}^3 \mid 2x = 1, y = z\}$

(C) $\{(x, y, z) \in \mathbb{R}^3 \mid x \geq 0\}$

(D) $\{(x, y, z) \in \mathbb{R}^3 \mid x^2 = y^2\}$

(E) $\{(x, y, z) \in \mathbb{R}^3 \mid x = y = z\}$

Question 19 **Points: 1**

Given is an arbitrary rotation $R : \mathbb{R}^2 \to \mathbb{R}^2$ around the origin, and an arbitrary reflection $T : \mathbb{R}^2 \to \mathbb{R}^2$ across a line through the origin. Which of the following statements are true in general? One or multiple options can be correct.

(A) $RTRT = I$, where I is the identity.

(B) $R^2 = I$, where I is the identity.

(C) TRT is a rotation.

(D) $T(x) = x$ for infinitely many points $x \in \mathbb{R}^2$.

(E) $R(x) \neq x$ for all $x \in \mathbb{R}^2$.

Question 20 **Points: 1**

Given is the matrix $M = \begin{bmatrix} 3 & 3 & 0 \\ 0 & -1 & 0 \\ 2 & 0 & 2 \end{bmatrix}$ over the finite field $\mathbb{Z}_5$. Which of the following are eigenvalues of M? One or multiple options can be correct.

(A) 1

(B) 2

(C) 3

(D) 4

(E) 0

Question 21 **Points: 1**

Which of the following are true statements about the matrix $M = \begin{bmatrix} 1 & 5 \\ 3 & 3 \end{bmatrix}$ with entries in the finite field $\mathbb{Z}_7$? One or multiple options can be correct.

(A) The determinant of M is 2.

(B) 6 is an eigenvalue of M with corresponding eigenspace $\left\{ \begin{bmatrix} a \\ a \end{bmatrix} : a \in \mathbb{Z}_7 \right\}$.

(C) M has two distinct eigenvalues in $\mathbb{Z}_7$.

(D) The rank of M is 1.

(E) $\det(M) = \det(M^{-1})$.

Question 22 **Points: 1**

Which of the following products of permutations are even? One or multiple options can be correct.

(A) 1

(B) (1 2 3 4 5)

(C) (1 2) (3 2 5)

(D) (1 2 3) (4 1 2 5) (3 4)

(E) (1 2 3 4) (3 2 5)

Question 23 **Points: 1**

A square matrix M is said to be *idempotent*, if $M^2 = M$. Which of the following matrices are idempotent over the specified fields? One or multiple options can be correct.

(A) $\begin{bmatrix} 5 & 2 \\ -10 & -4 \end{bmatrix}$ with entries in $\mathbb{R}$

(B) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ with entries in $\mathbb{Z}_{11}$

(C) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with entries in $\mathbb{Z}_5$

(D) $\begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$ with entries in $\mathbb{Z}_5$

(E) $\begin{bmatrix} 4 & -4 \\ 3 & -3 \end{bmatrix}$ with entries in $\mathbb{R}$

Question 24 **Points: 1**

Which of the following sets of vectors form bases of $\mathbb{Z}_2^3$? One or multiple options can be correct.

(A) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

(B) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$

(C) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$

(D) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

(E) $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

Question 25 **Points: 1**

Consider the subspace

$$W = \left\{ \begin{bmatrix} a \\ 3a+b \\ 2b \\ 0 \end{bmatrix} : a,\ b \in \mathbb{R} \right\}$$

of $\mathbb{R}^4$. Which of the following are spanning sets for W? One or multiple options can be correct.

(A) $\left\{ \begin{bmatrix} 2 \\ 6 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} \right\}$

(B) $\left\{ \begin{bmatrix} 1 \\ 3 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 2 \\ 0 \end{bmatrix} \right\}$

(C) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \\ 0 \end{bmatrix} \right\}$

(D) $\left\{ \begin{bmatrix} 0 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 0 \\ 0 \end{bmatrix} \right\}$

(E) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Question 26 **Points: 1**

Which of the following sets of vectors are linearly independent in the specified vector space? One or multiple options can be correct.

(A) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \right\} \in \mathbb{Z}_3^3$

(B) $\left\{1, x, x^3\right\} \in \mathbb{P}_4$

(C) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right\} \in \mathbb{R}^4$

(D) $\left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right\} \in \mathbb{Z}_5^2$

(E) $\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\} \in \mathbb{Z}_3^2$

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Question 27 **Points: 1**

Which of the below pairs of functions are solutions to the following system of differential equations, where $x = x(t)$ and $y = y(t)$ are differentiable functions of a real variable t:

$$\begin{array}{rcrcr} x' & = & x & + & 3y \\ y' & = & & & y \end{array}$$

One or multiple options can be correct.

(A) $x(t) = te^t$ and $y(t) = \frac{1}{3}e^t$

(B) $x(t) = (2+t)e^t$ and $y(t) = e^t$

(C) $x(t) = 3(1+t)e^t$ and $y(t) = e^t$

(D) $x(t) = e^t$ and $y(t) = 1$

(E) $x(t) = 10e^t + 6te^t$ and $y(t) = 6e^t + 10te^t$

Question 28 **Points: 1**

let $\langle -, - \rangle : V \times V \to \mathbb{R}$ be an inner product on vector space V over the real numbers $\mathbb{R}$. Which of the following statements are necessarily true? One or multiple options can be correct.

(A) $\langle u, v \rangle = \langle v, u \rangle, \quad \forall u, v \in V$

(B) $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle, \quad \forall u, v, w \in V$

(C) $\langle u + v, w + x \rangle = \langle u, w \rangle + \langle v, x \rangle, \quad \forall u, v, w, x \in V$

(D) $\langle u, v \rangle = -\langle v, u \rangle, \quad \forall u, v \in V$

(E) $\langle u, v \rangle \leq \sqrt{\langle u, u \rangle} \cdot \sqrt{\langle v, v \rangle}, \quad \forall u, v \in V$

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Question 29 **Points: 1**

Which of the following pairs of vectors complete $\left\{ \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} \right\}$ to an orthonormal basis of $\mathbb{R}^4$? One or multiple options can be correct.

(A) $\left\{ \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

(B) $\left\{ \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ -2 \\ 1 \\ 2 \end{bmatrix}, \frac{1}{\sqrt{15}} \begin{bmatrix} -1 \\ 2 \\ -1 \\ 3 \end{bmatrix} \right\}$

(C) $\left\{ \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

(D) $\left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \frac{1}{2} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$

(E) $\left\{ \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$

This is the end of the examination paper

This final exam comprises seven questions. Upload your solutions as an assignment in canvas, through Gradescope. Please write carefully and legibly. In order to get full credit, ***show all working****, unless the question specifies otherwise. Use the notation introduced in lectures. Present your arguments clearly,* ***using words of explanation*** *that connect formal use of symbols.*

After you upload your solutions, Gradescope will prompt you to ***tag the different problems****. It is your responsibility to do so correctly. If you do not, problems may go unmarked. Remember that technical or internet issues can always happen, so do not leave scanning, uploading and tagging until the very last moment.*

1. For each of the following statements, indicate whether they are true or false. If a statement is true, prove it. If a statement is false, give a counterexample.

   (a) Every nonzero element in $\mathbb{Z}_9$ has a multiplicative inverse.

   (b) For every prime number p, the matrix $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ over $\mathbb{Z}_p$ is invertible.

   (c) Any square matrix with characteristic polynomial $(\lambda - 1)(\lambda - 2)(\lambda - 3)^2$ is not diagonalisable.

   (d) Every doubly stochastic matrix has eigenvector $\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$. *(Recall that a double stochastic matrix is a square matrix whose rows and columns are probability vectors).*

   (e) Suppose that $(x, y) \in \{(1, 2), (2, 1)\}$. Then the configuration

x	3	y
6		8
7	5	4

   can be reached from the 8-puzzle

1	2	3
4	5	6
7	8	

   by moving squares in and out of the space.

   (f) Let A, P be real matrices. If P is invertible and $B = PAP^{-1}$, then $e^B = Pe^AP^{-1}$.

2. Consider $\mathbf{v_1} = (1,1,1)$, $\mathbf{v_2} = (1,2,3)$ and $\mathbf{v_3} = (0,1,4)$ in $\mathbb{Z}_5^3$.

   (a) Verify that $\mathbf{v_1}$, $\mathbf{v_2}$ and $\mathbf{v_3}$ are linearly independent in $\mathbb{Z}_5^3$.

   (b) Find the coordinate vector of $(3,2,2) \in \mathbb{Z}_5^3$ with respect to the basis $\{\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}\}$.

3. Let n be a positive integer and F a field. Recall that we write $\mathrm{GL}_n(F)$ for the group of invertible $n \times n$ matrices over F under multiplication.

   (a) Prove that
   $$H = \{A \in \mathrm{Mat}_n(F) \mid A^T A = I\}$$
   is a subgroup of $\mathrm{GL}_n(F)$.

   (b) Now suppose $F = \mathbb{R}$ and $A \in \mathrm{Mat}_n(\mathbb{R})$. Prove that $A \in H$ if and only if the columns of A form an orthonormal set in $\mathbb{R}^n$.

4. Let B be the real matrix
   $$B = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 3 & 0 \\ -2 & 0 & 3 \end{bmatrix}.$$

   (a) Find an invertible matrix P and a diagonal matrix D such that $B = PDP^{-1}$.

   (b) Solve the following system of differential equations,
   $$\begin{array}{rcrcr} x' & = & x & & \\ y' & = & -2x & + & 3y \\ z' & = & -2x & + & 3z \end{array}$$
   where $x = x(t)$, $y = y(t)$ and $z = z(t)$ are differentiable functions of a real variable t, with initial conditions $x(0) = 1$, $y(0) = -1$ and $z(0) = 0$.

5. The real matrix A has reduced row echelon form R, where
   $$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 3 & -1 \\ 1 & 2 & 5 & -4 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$
   Let $W \subset \mathbb{R}^4$ be the row space of A.

   (a) Give a basis for W.

   (b) What is the dimension of W?

   (c) What is the nullity of A?

   (d) Apply the Gram-Schmidt process to find an orthonormal basis for W.

6. Consider the basis $\mathcal{B} = \{1,\, x,\, x^2\}$ for $\mathbb{P}_2$.

(a) Suppose the coordinate vector of $f(x) \in \mathbb{P}_2$ with respect to $\mathcal{B}$ is $[f(x)]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$.

Write $f(x)$ as a polynomial. *You do not need to justify your answer for part (a).*

(b) Let $T : \mathbb{P}_2 \longrightarrow \mathrm{Mat}_2(\mathbb{R})$ be the linear transformation defined by

$$T\left(f(x)\right) = \begin{bmatrix} f(1) & 0 \\ f(5) & -f(1) \end{bmatrix}.$$

For example, $T(1 + x + x^2) = \begin{bmatrix} 3 & 0 \\ 31 & -3 \end{bmatrix}$.

Consider the basis

$$\mathcal{E} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

for $\mathrm{Mat}_2(\mathbb{R})$.

Find the matrix of T with respect to $\mathcal{B}$ and $\mathcal{E}$, in other words find $[T]_{\mathcal{E}}^{\mathcal{B}}$.

7. Consider the stochastic matrix

$$C = \begin{bmatrix} 0.8 & 0.3 & 0.3 \\ 0.1 & 0.5 & 0.2 \\ 0.1 & 0.2 & 0.5 \end{bmatrix}.$$

(a) Find a steady state probability vector for C.

For part (b) and (c), you do not have to explain or show your work:

(b) Find the dominant eigenvalue of C.

(c) Find $\lim\limits_{n \to \infty} C^n$.